

Master Thesis — Progress Report

Spain's MTU15 reform: empirical assessment in progress

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Outline

- **1. Reminder.** Objective, research questions, context.
- **2. What I have done.** Data, methods, headline findings, tests of theory.
- **3. State of the work.** Claim portfolio, theoretical model.
- **4. Difficulties encountered.** Identification, an unexplained break.
- **5. What is left.**

Following advisor's guidance: a progress report rather than a results talk. 18–20 minutes.

Objective and concrete research questions

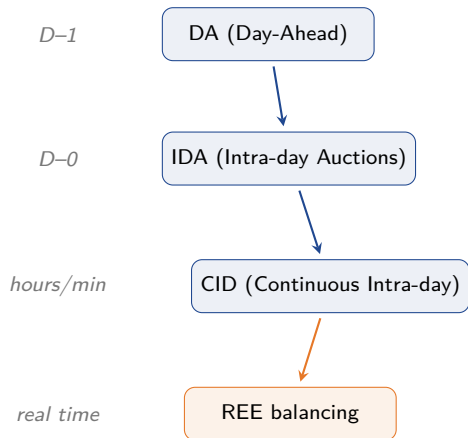
Aim

Empirically assess the impact of Spain's 2024–2025 **MTU15 reform sequence** on market outcomes and firm strategic behaviour, and extend the theoretical framework to capture the mechanism.

Five concrete sub-questions:

1. Does the **asymmetric-clock window** (DA60 + ID15 + ISP15) generate a measurable dual-pricing redistribution from BRPs to the TSO? Does it close at MTU15-DA?
2. Does finer market-time-unit granularity **reduce strategic forward-quantity withholding** (Allaz–Vila / Ito–Reguant prediction)?
3. Does the reform create **heterogeneous winners and losers** across firms and technologies?
4. Does the per-segment incidence of imbalance settlement match a **Pigouvian benchmark**?
5. Has *operación reforzada* (post-blackout regime) broken the separability of the reform sequence?

Big picture — D-1 to delivery



- **DA** (uniform-price auction): main stage, sets baseline schedule and price.
- **IDA** (uniform-price auctions): three discrete sessions to update positions as forecasts and availability change.
- **CID** (continuous matching, XBID/SIDC): pay-as-bid, last voluntary adjustment.
- **REE balancing + imbalance settlement**: residual deviations + system security; ex-post dual-pricing rule (EBGL Art. 52).

The 2024–2025 reform sequence

- Traditionally MTU was **60 min** everywhere.
- Three stages:
 - **Dec 2024** — **ISP15**: imbalance settlement → 15-min.
 - **Mar 2025** — **ID15**: intraday → 15-min.
 - **Oct 2025** — **DA15**: day-ahead → 15-min.
- Two transitional regimes where *settlement* and *trading* clocks differ.
- Plus: post-2025-04-28 blackout ⇒ *operación reforzada* (REE's reinforced-operation regime), continuous since.

Period	ISP	ID	DA
Pre-Dec 2024	60	60	60
Dec24–Mar25	15	60	60
Mar25–Sep25	15	15	60
Oct25 onward	15	15	15

ISP15

Dec 2024

ID15

Mar 2025

DA15

Oct 2025

Data pipeline (1/2) — what each layer lets us see

Answering the five research questions requires reconstructing the *entire trace* of an electricity transaction — from a firm's first bid in the day-ahead auction to the final cash flow at imbalance settlement. So I built the data infrastructure end-to-end.

To see prices and the reform itself — OMIE (Iberian market operator)

Day-ahead clearing prices, intraday auction prices (per session), and the entire continuous-market price record at native granularity (MTU60 pre-2025-03-19, MTU15 after). This is what tells me *whether* the reform moved equilibrium prices, when, and by how much.

To see strategic conduct — OMIE bid-level data

Every tranche of every offer ever submitted to the day-ahead and IDA auctions, plus the full XBID order-book and matched-trade record. This is what lets me test Allaz-Vila / Ito-Reguant / Hortaçsu-Puller predictions at the level the theory speaks — the individual bid function.

To see imbalance and the system response — ENTSO-E + REE/ESIOS

ENTSO-E TP: imbalance volumes and prices, FRR settlement (A87), generation by type and unit, load and renewable forecasts. All fundamentals controls and forecast errors come from here.

REE/ESIOS: per-firm and per-segment settlement, real-time RT prices, balancing offers.

Caveat: ESIOS API access still pending — 3-week wait. Public archive endpoints work; per-BRP feeds blocked until the token clears.

Data pipeline (2/2) — the dimension of what we have

Putting it together, the panel is unusually rich for a Master's thesis. The whole point is that strategic mechanisms operate at sub-hourly granularity — aggregating ex ante would erase exactly what we are trying to identify.

Scale

- ~**8 years** of coverage (2018-01 → 2026-02), at native MTU60/MTU15 granularity throughout.
- ~**413 GB raw** + ~**198 GB processed** on dedicated external SSD; 25+ data families with their own parsers.
- ~**5.7 billion rows** on the OMIE side alone, headlined by:
 - Continuous-market final hourly programs (PHFC): **3.3 billion rows**.
 - IDA bid tranches (IDET): 314M rows. DA bid tranches (DET): 172M rows.
 - Day-ahead bilateral programs (PDBF): 166M rows — recently added; opens the bilateral-channel mechanism.
 - Aggregate DA supply–demand curves: 341M rows.
 - XBID order book: 163M orders, 32M matched trades.

Why it matters for identification

Most papers that study Spanish wholesale markets rely on *published* hourly price series (~70k rows). Working at the bid-tranche / firm-ISP level lets me cluster at the right grain (week-of-sample, day, or unit), control for forecast errors at native frequency (Fabra–Imelda style), and run robustness checks on samples in the millions of observations rather than hundreds. This is where the project's empirical credibility lives.

Empirical strategy

- **No clean external counterfactual** — most comparable EU markets adopted MTU15 already and differ structurally.
- Reduced-form before/after with rich controls:
 - Fundamentals: load, wind + solar (actual + forecast errors), fuel/CO₂.
 - Calendar: cal-month FE, day-of-week, hour-of-day; year FE where appropriate.
 - Reform-date and *reforzada* indicators.
- **Same-calendar-month robustness** mandatory for any cross-regime claim (Spain is highly seasonal).
- **Cluster-robust SE** by date or week-of-sample (Cameron–Miller $G \geq 30$ rule).
- **Cross-country placebo**: France DA prices flat across Spanish reform dates.
- **Structural-time-series counterfactual** (Brodersen et al. 2015): ongoing.

Headline findings

1. System-layer settlement transfer

During the asymmetric-granularity window (Dec 2024–Sep 2025) the BRP→TSO net transfer was **+€1,094.9M cumulative** vs same-calendar pre-reform baseline (bootstrap CI $[-90, +73]$ M $\Rightarrow \approx 15\times$ upper bound). Collapses to **€7.4M/mo at MTU15-DA** ($12\times$ drop). Regulatory *redistribution*, not deadweight.

2. Mechanism — forecast-error pass-through to imbalance

Pooled regression, daily grain: net per-regime slope β is ~ 0 pre-reform, ~ 0 at 3-session IDA, -0.03 at ISP15-win, **+0.09 at DA60/ID15** (uniquely elevated), ~ 0 at DA15/ID15. Robust to year FE.

3. Firm-level market power (Iberdrola)

Synthetic-firm replication à la Ciarreta–Espinosa: **Iberdrola alone** $\sim$ **€820M** day-ahead cleared-price-difference rent post-MTU15-IDA (98% of joint Big-4); hydro 64% / CCGT 36%. **Regime-invariant**. Behavioural complement: Big-4 voluntary IDA repositioning (q_2 , à la Ito–Reguant) collapses progressively, recovers at DA15/ID15; robust under canonical Fabra–Imelda controls.

Tests of theoretical predictions — with the literature

What worked:

- **Ito & Reguant (2016, AER)** sequential markets, market power, arbitrage: predicted q_2 collapse if firms react to a finer spot grid → *confirmed* (regime coefficients vary $< 0.1\%$ across spec battery).
- **Ciarreta & Espinosa (2010, J Regul Econ)** synthetic-firm method → *replicated and extended* (€820M IB rent; survives sensitivity tests).

What did not work:

- **Allaz & Vila (1993, JET)** forward markets dispel rents: my test was the slope $\partial \Delta Q_{IDA} / \partial q_{DA}$ → *rejected as a test* ($Q_{actual} \approx q_{DA} + \Delta Q_{IDA}$ is a mechanical accounting identity, not a structural slope). The surviving Allaz–Vila evidence comes from the Ito–Reguant q_2 trajectory above.
- **Hortaçsu & Puller (2008, RAND)** sophistication: implied Cournot–FOC Lerner vs realised marginal-bid markup → *strategic-conduct interpretation rejected* (implied ≈ 0.93 vs realised ≈ 0.34 for the leading firm; implied is formula-mechanical).
- **Bushnell (2003, JIE)** hydro market power via reservoir shadow-price → *mechanism rejected* (no reservoir gradient; placebo stronger than treatment). The IB hydro Q4 concentration is real but not classical Bushnell.

State of the work

Empirical portfolio:

- I track findings formally with a status (alive, wounded, dead) and a documented robustness trail.
- Currently: ~40 alive, ~7 wounded, ~20 dead (audit trail).
- Each cited number is regression-backed at proper grain and survives multi-spec robustness.

Three policy levers identified:

1. Clock-symmetry — *implemented* at MTU15-DA.
2. Pigouvian incidence — *open*.
3. Operational regime overlay (*operación reforzada*) — *open*.

Theoretical model:

- Started from an extension of Ito–Reguant (2016) with explicit MTU mismatch (in my Energy Economics proposal).
- Empirically the three predictions hold; magnitudes are larger than the theory suggested.
- **Pivot:** *very solid empirics* + a *simplified IO model* for mechanism + a light first-order-condition exercise.
- **Not** a fully estimated structural auction model.

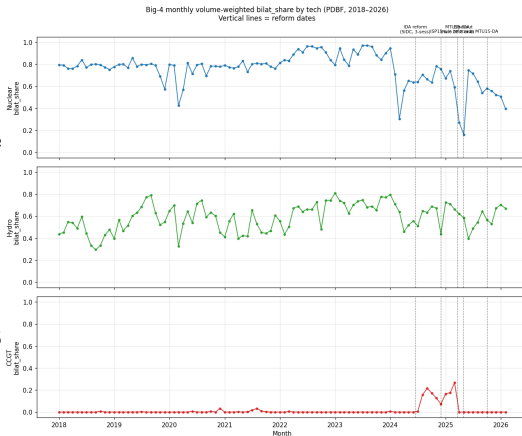
Difficulty 1 — identification under multiple confounds

- Each reform date has at least one competing event:
 - *Dec 2024*: European gas-price normalisation.
 - *Mar 2025*: CNMC eliminates the bilateral-contract opportunity-cost rule simultaneously.
 - *Apr 28 2025* blackout → *operación reforzada* (regime overlay continuous since).
 - *Oct 2025*: partial overlap with reforzada.
- **Strategy adopted:**
 - *Market-chain placement*: reforzada acts *after* the day-ahead clearing ⇒ DA-side outcomes face it only via expectations.
 - *Reforzada-held-constant pairings*: ISP15-win → DA60/ID15-PRE-blackout (intraday reform); DA60/ID15-POST → DA15/ID15 (DA reform).
 - Same-cal-month + cross-country placebo + bootstrap CIs.
- Honest record of **14 dead causal-identification attempts** (DiD, wind-IV, profile-matching) — rejected after placebo tests.

Difficulty 2 — a structural break in Q1 2024 we cannot yet explain

- Big-4 **nuclear bilateral-contract share** stable at 80–95% across 2018–2023.
- Drops sharply in **Feb–Mar 2024**, three months *before* any MTU15 reform.
- Hydro shows the same Q1-2024 break (auction-side growth); CCGT unaffected.
- Concurrent with the **end of the Iberian gas-cap (Dec 2023)** and the first major renewable-saturation event in Spain (negative DA prices appear Apr 2024).
- Regression on negative-price-share (exogenous proxy, with unit/cal-month FE): nuclear $\beta = -3.05$ ($R^2 = 0.13$); the implied magnitude matches the observed drop. Hydro $\beta \approx 0$.

Mechanism unidentified; reframes how I read 2024 pre-reform data.



What is left

Empirical:

- Pin down the Q1 2024 mechanism (gas-cap / renewables / PPAs).
- Robustness re-pass on remaining alive findings (per-firm Lerner, bid complexification, post-blackout CCGT/aFRR).
- Brodersen-style structural-time-series counterfactual on headline outcomes.
- Per-segment settlement verification once the ESIO API token clears.

Theoretical & writing:

- Refine the Ito–Reguant extension to motivate the three-lever framing.
- Light HP-style FOC exercise on Big-4 bid functions; no full structural estimation.
- Drafting starts the week after this talk; deadline mid-June 2026.
- **Risk:** 8-week timeline is tight; firm-level Part II is the binding constraint.

Thank you.

Questions and feedback welcome.

Pablo Paramio -- CEMFI -- May 2026